

ENTERPRISE SECURITY ASSESSMENT LAB

AD NETWORK SECURITY METHODOLOGY REPORT

Phase 1 - Establishing AD reachability and mapping domain services before
credential-based testing

Report Section	Phase 01 - Lab Connectivity and Domain Enumeration
Project Area	Internal Network and Active Directory Security Assessment
Environment	Private Windows AD lab - Kali Linux, Windows Server, Windows endpoint
Prepared For	GitHub Portfolio Documentation
Document Type	Individual methodology PDF with redacted evidence

Authorized lab documentation only. Sensitive credential material, hashes, passwords, and replication output have been redacted before publication.

1. Executive Summary

This report documents **Lab Connectivity and Domain Enumeration** as part of the AD Network module in the Enterprise Security Assessment Lab. The section is written as a professional, portfolio-safe methodology report that explains the testing objective, workflow, evidence, tool usage, sanitized output, security risk, remediation guidance, and GitHub publication requirements.

Objective: Confirm that the private Active Directory lab is reachable, identify the Domain Controller, and verify that expected AD services such as DNS, Kerberos, LDAP, SMB, and Global Catalog are exposed before continuing with credential-based testing.

MITRE / Framework Mapping	Discovery and Service Enumeration - mapped conceptually to Remote System Discovery (T1018) and Account/Domain Discovery practices
Testing Scope	Authorized private Active Directory lab only
Evidence Handling	Screenshots embedded with necessary redaction; raw hashes and secrets are not included

2. Methodology and Workflow

The methodology below is designed to show a controlled penetration-testing workflow rather than random tool execution. Each step supports a specific assessment goal and produces evidence that can be reviewed safely in a public portfolio.

- Verify Kali has an interface connected to the AD lab network.
- Confirm basic reachability to the Domain Controller.
- Run host discovery to identify live machines.
- Scan only expected AD service ports first to keep the methodology controlled.
- Document the results as the baseline for later credential, Kerberos, SMB, and BloodHound testing.

Command / Workflow Used

```
# Identify local network interfaces
ip -br a

# Discover live systems in the lab subnet
nmap -sn <LAB_SUBNET>

# Scan common Domain Controller services
nmap -sV -sC -p 53,88,135,139,389,445,464,636,3268,3389 <DC_IP>
```

Sanitized Output Style

```
Expected AD services on a Domain Controller:
53/tcp DNS
88/tcp Kerberos
135/tcp MSRPC
139/tcp NetBIOS
389/tcp LDAP
445/tcp SMB
464/tcp Kerberos password change
636/tcp LDAPS
3268/tcp Global Catalog
3389/tcp RDP (if enabled)
```

3. Evidence Mapping and Screenshot Explanation

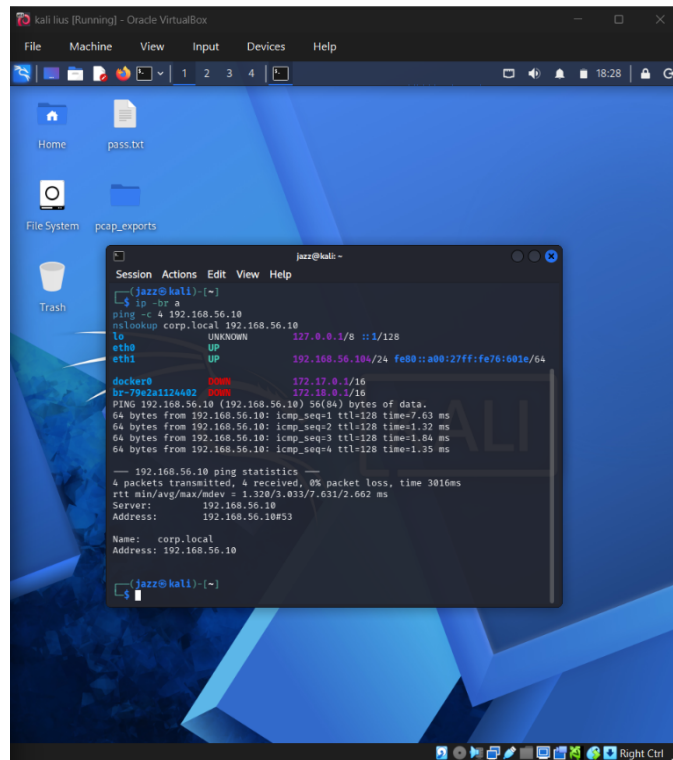


Figure - Kali connectivity to the Domain Controller

This screenshot proves that the attack/testing machine can reach the Domain Controller over the private lab network.

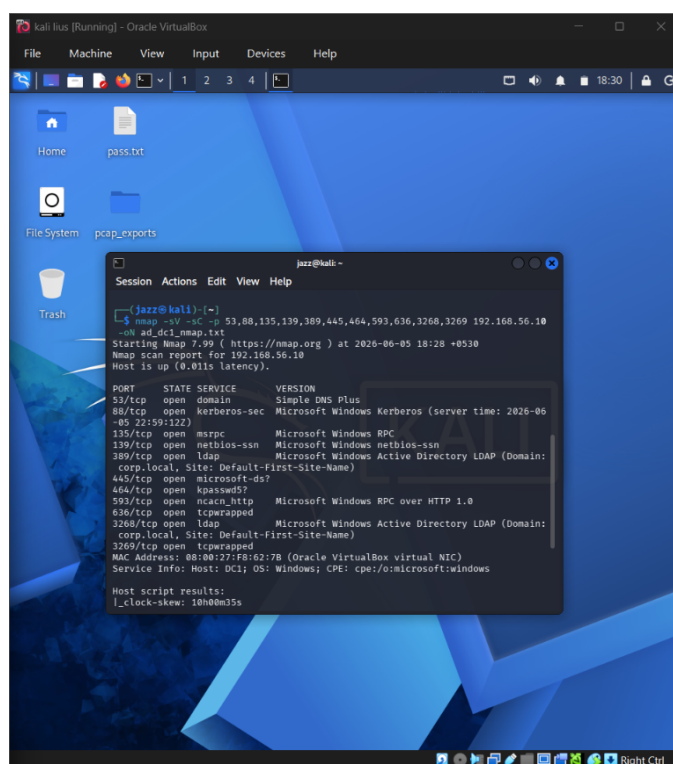


Figure - Nmap scan of Active Directory service ports

This screenshot demonstrates validation of AD-relevant ports such as DNS, Kerberos, LDAP, SMB, and Global Catalog.

- Segment workstations and servers where possible.
- Monitor LDAP, SMB, Kerberos, and RDP access patterns.
- Disable unnecessary services and review firewall rules.
- Maintain asset inventory and regularly compare expected services with actual exposure.

7. GitHub Redaction and Publication Guidance

- Publish only redacted evidence. Do not upload raw hashes, password material, ticket files, DCSync output, or secretsdump text dumps.
- Keep this report under 03-ad-network/reports/methodology/ and keep raw tool output in a private working folder unless fully sanitized.
- Use placeholders such as , , , and in public documentation.
- Confirm the screenshot file name in the README matches the repository file name exactly, including case, hyphens, and extension.

Do Not Upload

```
*.hash
*_hashes.txt
responder logs with raw NetNTLMv2
kerberoast_hashes.txt
secretsdump output
DCSync dumps
cracked-password files
unredacted BloodHound collection ZIPs if they expose sensitive names
raw command history containing passwords
```

8. Conclusion

The **Lab Connectivity and Domain Enumeration** phase demonstrates a structured and evidence-based AD testing workflow. The report explains the purpose of the test, shows sanitized evidence, connects the activity to real-world risk, and provides practical remediation guidance. This makes it suitable for a professional cybersecurity GitHub portfolio.